Lesson plan · for the teacher

Models and predictions: unit check

Data science · Level 3 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Lines that predict
- Training and testing
- Nearest neighbours
- Judging a model

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet for its answers.
2. Quiz (20 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask a team to explain it with an example from the unit.
4. Wrap-up (8 min): Work through the question most teams got wrong, together on the board.

Discussion

- Which idea will you use first?
- Which school model would you question?
- Which mistake costs most at your school?

Exit ticket

- One prediction my class could test fairly this week.

Try it in class tomorrow
Run the hidden-cards activity with a class, and ask them to judge each rule by its test error, not its training error.

Worksheet · for pupils

Models and predictions: unit check

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Name _____ Class _____ Date _____

1 **Use the line.**

mark = 6 + 3 × hours, so 2 hours predicts _____ marks.

Word bank: 12 · 9

2 **A line fitted to 0–5 hours can be trusted at 40 hours.**

◯ True ◯ False

3 **Which score is an honest estimate for new pupils?**

◯ The test score ◯ The training score ◯ The best score ever seen

4 **Put the steps in order.**

☐ Test once on the test set

☐ Train on the training set

☐ Split the data

5 **Match each term to its meaning.**

Overfitting	• • • •	The score above which the model says yes
Classification	• • • •	Predicts a label
Threshold	• • • •	Fits the noise in training data